



EDUCATION

Virginia Polytechnic Institute and State University: Blacksburg, VA **2025-2028**

- General Engineering; Prospective Major: Robotics and Mechatronics in Mechanical Engineering

Ocean Lakes High School and Advanced Technology Center: Virginia Beach, VA **2021-2025**

- 3.99 Unweighted, 4.616 Weighted GPA; Graduate of The Math and Science Academy
- Engineering Technology Program (2022-23), Engineering Practicum (2024)

ENGINEERING EXPERIENCE

MachWorks Engineering Design Team **2025-Present**

- Testing and experimenting with Kalman filters to process IMU and GPS data from UAV and running computer vision models on a Jetson Xavier.

Underwater ROV and Telemetry System **2024-2025**

- Designed ROV out of PVC and bilge pumps; Hijacked PS2 controller to send controls to Arduino utilizing RS485 protocols; Created CAD models of the project and technical documentation

Technology Student Association **2023-2024**

- **CAD Engineering:** Utilized provided isometric drawings to create CAD models, technical drawings, and renders under time constraint (Competed at regionals, states, and nationals)
- **Flight Endurance:** Designed a rubberband powered glider kit aimed at longest airtime (Competed at regionals and states)
- **Webmaster:** Designed website with teammates for hypothetical business model (Competed at regionals, states, and nationals)

American Society of Materials Eisenman Delegate **Summer 2024**

- 1/30 nationally who earned a spot in the camp after winning ASM Materials challenge
 - Built a device to test lower range tensile strength values of materials
- Performed failure analysis on stud and utilized SEM, EDS, and analyzed the carbon content

SKILLS

- **Certifications:** Certified SolidWorks Associate and Professional (CSWP) in Mechanical Design
- **Skills:** SolidWorks models, assemblies, and drawings, Vectric CNC, Adobe Illustrator, 3D Printing, Technical Report Writing, Laser Cutting, Soldering, Prototyping

LEADERSHIP

ABRC Manager (23-25) and Lifeguard (22-23) **2022-2025**

- Summer employment managing lifeguard staff, maintaining pool chemistry and equipment, and providing customer service and support

Bolts Volunteer Swim Coach **2020-2025**

- Taught children ages 4-12 how to swim and how to improve their stroke technique

ODS Middle School Leadership Workshop Staff (22-24) and Coordinator (24-25) **2022-2025**

- Created interactive curriculum; was responsible for logistics with ~ 60 staff and 200 students

Executive Board for OL Class of 2025 **2022-2024**

- Organized and promoted class socials and helped raise ~\$15,000 for Ring Dance and Prom

National Technical Honor Society **2022-2024**

- Leader in Safety Safari Program teaching local elementary schoolers internet safety
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